

Name



ID#

Math 150 Spring 2025 Final Exam

- Desmos is allowed (and the only app website allowed - no other apps or websites are allowed at anytime)
- Must show all work to receive credit
- All integrals must be
- Attach all scratch paper

	Question	Points
1	Integration	1 / 5
2	Riemann Sums	2 / 5
3	Volume	1 / 5
4	Limits & Continuity	3 / 4
5	Tangent & Normal Lines	0 / 4
6	Surface Area*	2 / 4
7	Optimization	0 / 4
8	Related Rates	1 / 3
9	First Derivative Test	1 / 3
10	Geometry	2 / 3
Total		11 / 40

1) Evaluate the integral $\int 43 \tan^5(x) \sec^4(x) dx$ using two different substitutions and discuss how the two anti-derivatives are related.

$$43 \tan^5(x) \sec^4$$

$$u = \sec$$

$$du = \sec \tan$$

$$13 \int \tan^4 \sec^2 x + \tan^3 x \sec^2 x$$

$$\int \sec(x) - 1 \tan^3 x + \tan^3 x \tan$$

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2) Estimate the signed area under the curve $y = \sin\left(\frac{\pi x}{4}\right)$ from -2 to 4 using:

A B
[-2, 4]

(a) three (3) rectangles with the **left endpoint** approximation.

(b) three (3) rectangles with the **right endpoint** approximation.

(c) three (3) rectangles with the **midpoint** approximation.

(d) Find the exact signed area using FTC and compare your estimates from parts (a-c)

In each case a-c draw the graph and rectangles and use sigma notation.

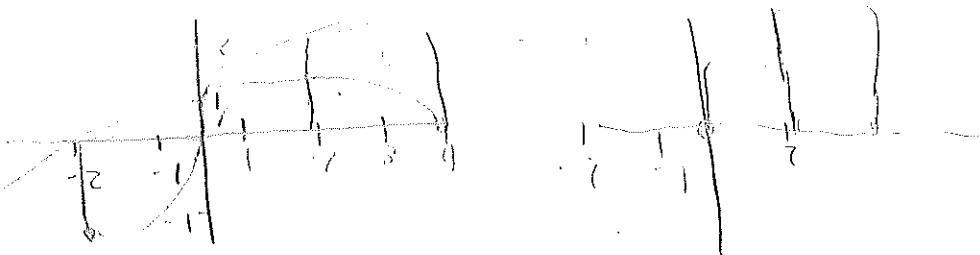
B

$$\int_{-2}^4 \sin\left(\frac{\pi x}{4}\right) dx$$

A X : 2 4 - (-2) = 6 $\frac{6}{3} = 2$

CFI

$$\begin{aligned} X_0 &= -2 + 0(2) = -2 \\ X_1 &= -2 + 1(2) = 0 \\ X_2 &= -2 + 2(2) = 2 \\ X_3 &= -2 + 3(2) = 4 \end{aligned}$$



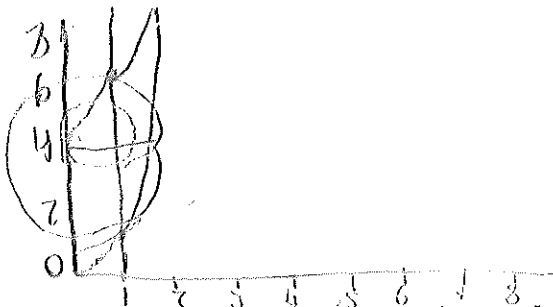
2X19 = 1

3) Find the volume generated by revolving the region bounded by the curves

$$y = 2x + 4, x = 1, y = x^3$$

about the y -axis first using washers, then using shells.

Draw the region and a typical washer. Draw the region and a typical shell.



Find the exact value of volume using washers.

$$\int_0^8 \pi (x^2 - 1) dx$$

$$\int_0^8 \pi (x^2 - 1) dx$$

Find the exact value of volume using shells.

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4) Sketch the graph of a function with following limits, and classify any discontinuities.

• $\lim_{x \rightarrow -2^-} f(x) = -2$ ✓ $(-\infty, -2)$

• $\lim_{x \rightarrow -2^+} f(x) = 1$ ✓ $(-2, 1)$

• $f(-2) = 3$ ✓ $(-2, 3)$

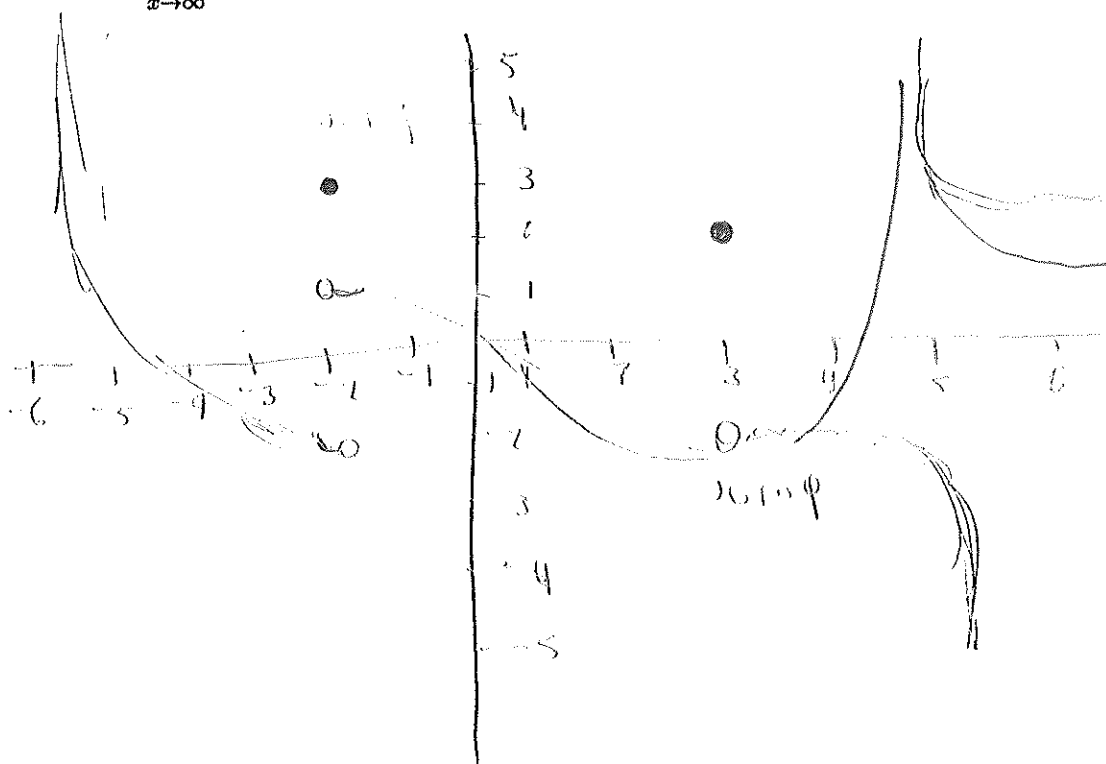
• $\lim_{x \rightarrow 3} f(x) = -2$ $(3, -\infty)$

• $f(3) = 2$ ✗ $(3, \infty)$

• $\lim_{x \rightarrow 5^+} f(x) = \infty$

• $\lim_{x \rightarrow 5^-} f(x) = \infty$

• $\lim_{x \rightarrow \infty} f(x) = 1$



5) Graph the equation $12(x^2 + y^2)^2 = 169xy^2$

Find the tangent line and normal line at the point (3, 2)

$$y - y_1 = m(x - x_1) \quad \text{Differentiate}$$

$$y - 2 = m(x - 3) \quad 12(x^2 + y^2)^2$$

$$12(x^2 + y^2) (2x + 2y)$$

$$(24x^2 + 24y^2) 2x + 2y$$

$$48x^3 + 48y^3$$

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6) Calculate the surface area of the surface obtained by revolving the curve $y = \frac{x^3}{3}$ around the x -axis from $1 \leq x \leq 2$.

$$2\pi \int_1^2 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\left\{ 2\pi \int_1^2 x^3 \sqrt{1 + (x^2)^2} = 73.4796 \right\}$$

$$F = x^3 \quad (F'(3) \cdot (F)(3)) = \frac{(3x^2)(3) - (x^3)(0)}{3^2} = \frac{9x^2}{3^2}$$

$$F' = 3x^2$$

$$f(3) = 3$$

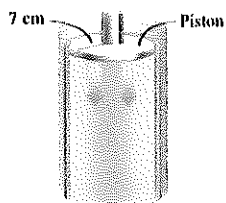
$$f'(3) = 0$$

$$2\pi$$

7) A wire of length 100 mm is cut into two pieces, forming a circle with the first segment, and a square with the remaining segment, while optimizing the sum of the area of the circle and square. State the objective function and the interval of interest.

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8) A piston is seated at the top of a cylindrical chamber with radius 7cm when it starts moving into the chamber at a constant speed of 3 cm / sec. (see figure).

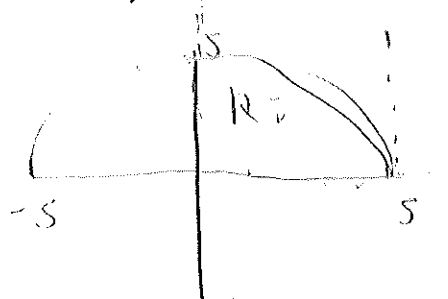


What is the rate of change of the volume of the cylinder when the piston is 9 cm from the base of the chamber?

$$\frac{dV}{dt} = 3 \quad r = 7$$

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10) Find the exact value of $\int_{-5}^0 \sqrt{25-x^2} dx$ using geometry.



Area of circle πr^2

$$\pi 5^2$$

$$\pi 25$$

$$\left[\frac{\pi 25}{4} \right] \checkmark$$

Find the exact value of $\int_{-5}^0 \sqrt{25-x^2} dx$ using integration techniques.

$$[-5, 0]$$

$$\sqrt{25-x^2}$$

$$0 - (-5) = 5$$

50 R.R.C

$$\Delta x = 10$$

50

$$\sum_{k=1}^5 -5 + (k-1) \frac{5}{50}$$

$$\frac{5k}{50} - \frac{5}{50} = \frac{5k-5}{50}$$

$$\sum_{k=1}^5 \sqrt{25 - \left(\frac{5k-5}{50} \right)^2} \quad 19.36$$

$$\frac{1}{\sqrt{25-x^2}} = \sin^{-1} \left(\frac{x}{5} \right) + C$$

$$\int \sin^{-1} \left(\frac{x}{5} \right)$$